

Name: _____

Find the slope of the line through each pair of points.

1. $(-3, 8)$ and $(5, -4)$

2. $(7, -2)$ and $(7, 6)$

Find the missing coordinate.

3. $m = 2/5$ through $(-1, 4)$ and $(9, \underline{\hspace{1cm}})$

4. $m = -3$ through $(\underline{\hspace{1cm}}, 1)$ and $(2, -8)$

Write an equation of each line in slope-intercept form. Check your answer.

5. through $(3, -2)$ and parallel to $y = -4x + 1$

6. through $(-6, 5)$ and perpendicular to $y = (2/3)x - 7$

7. through $(5, 1)$ and perpendicular to $x - 4y = 12$

Determine whether the lines are parallel, intersecting, or coinciding.

8. $y = 5x - 2$ and $10x - 2y = 4$

9. $y = -(1/2)x + 3$ and $2x - y = 6$

Solve the problem.

10. A wheelchair ramp rises 3 feet over a horizontal run of 24 feet. A second ramp must be parallel to the first and has a horizontal run of 40 feet. How many feet does the second ramp rise?